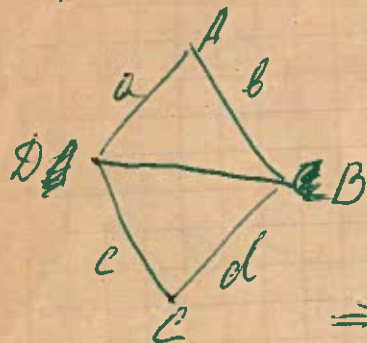


II remarque.



$$2S = ab \sin A + cd \sin C$$

$$4S^2 = a^2 b^2 \sin^2 A + c^2 d^2 \sin^2 C + 2abcd \cos(A-C) - abcd \cos(A+C)$$

$$DB^2 = a^2 + b^2 - 2ab \cos A = c^2 + d^2 - 2cd \cos C$$

$$\Rightarrow 2^2(ab \cos A - cd \cos C)^2 = (a^2 + b^2 - c^2 - d^2)^2$$

$$\left\{ \begin{aligned} (a^2 + b^2 - c^2 - d^2)^2 &= 4a^2 b^2 \cos^2 A + 4c^2 d^2 \cos^2 C - 4abcd \cos(A-C) - 4abcd \cos(A+C) \\ 16S^2 &= 4a^2 b^2 \sin^2 A + 4c^2 d^2 \sin^2 C + 4abcd \cos(A-C) - 4abcd \cos(A+C) \end{aligned} \right. \quad \left. \begin{array}{l} \text{cro-} \\ \text{num} \end{array} \right\}$$

$$(a^2 + b^2 - c^2 - d^2)^2 + 16S^2 = 4a^2 b^2 + 4c^2 d^2 - 8abcd \cos(A+C)$$

$$\Rightarrow S - \max \text{ lorsque } 8abcd \cos(A+C) \text{ minimum.}$$

$$\Rightarrow \cos(A+C) = -1 \quad \underline{A+C = \pi}$$